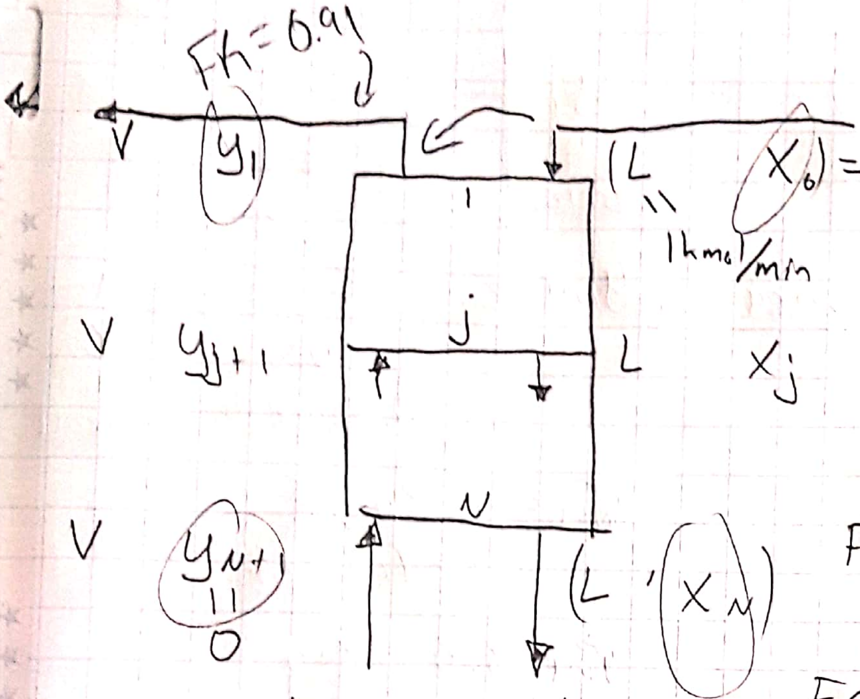
(2)

(1)

1) Stripper

90/100



$= 0.00012$

$T = 100°C$

$L/V = 20$

91% removed from liquid stream

$FR_{VB} = (1 - 0.91)$

Eq $y = 28x$
$y = m\,x$

a) $x_{out} = ?$ (x_N) $y_{out} = ?$ (y_1)

$$FR_{NB} = \frac{L X_N}{L X_0}$$

$$x_N = x_{out} = FR_{NB}\, X_0 = (1-0.91)(0.00012)$$

$$\boxed{x_N = out = 1.08 \times 10^{-5}}$$

MB: $V y_1 + L X_N = L x_0 + V \cancel{y_{N+1}}$ (0)

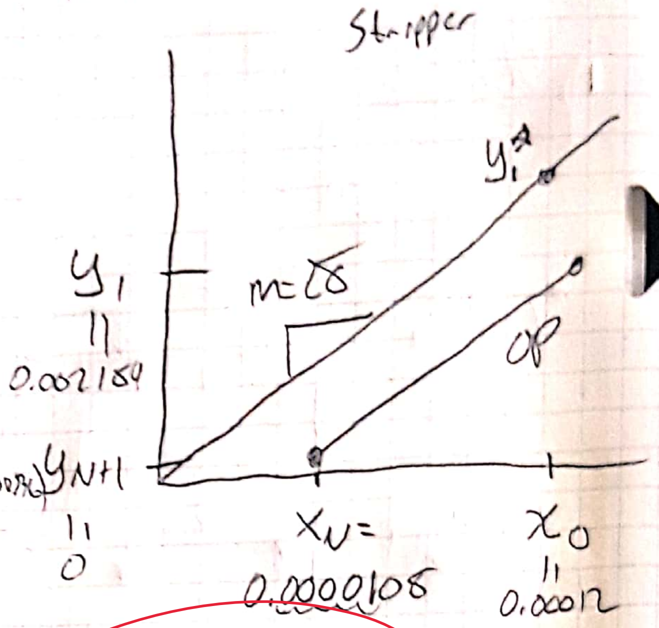
$$V y_1 = L x_0 - L X_N$$

$$y_1 = \frac{L}{V}(x_0 - X_N) = (20)(0.00012 - 1.08 \times 10^{-5})$$

$$\boxed{y_1 = 0.002184}$$

10/10

b) $N = ?$



$$N = \frac{\ln\left[\left(y_{N+1} - y^*_{N+1}\right)/\left(y_1 - y_1^*\right)\right]}{\ln\left(\frac{L}{V}\left(\frac{1}{m}\right)\right)}$$

$$= \frac{\ln\left[\left(-3.024\times10^{-4}\right)/\left(0.002154-0.00336\right)\right]}{\ln\left(20/28\right)}$$

$$\boxed{N = 4.036}$$

15/15

$y^*_{N+1} = m x_N + \cancel{b}$

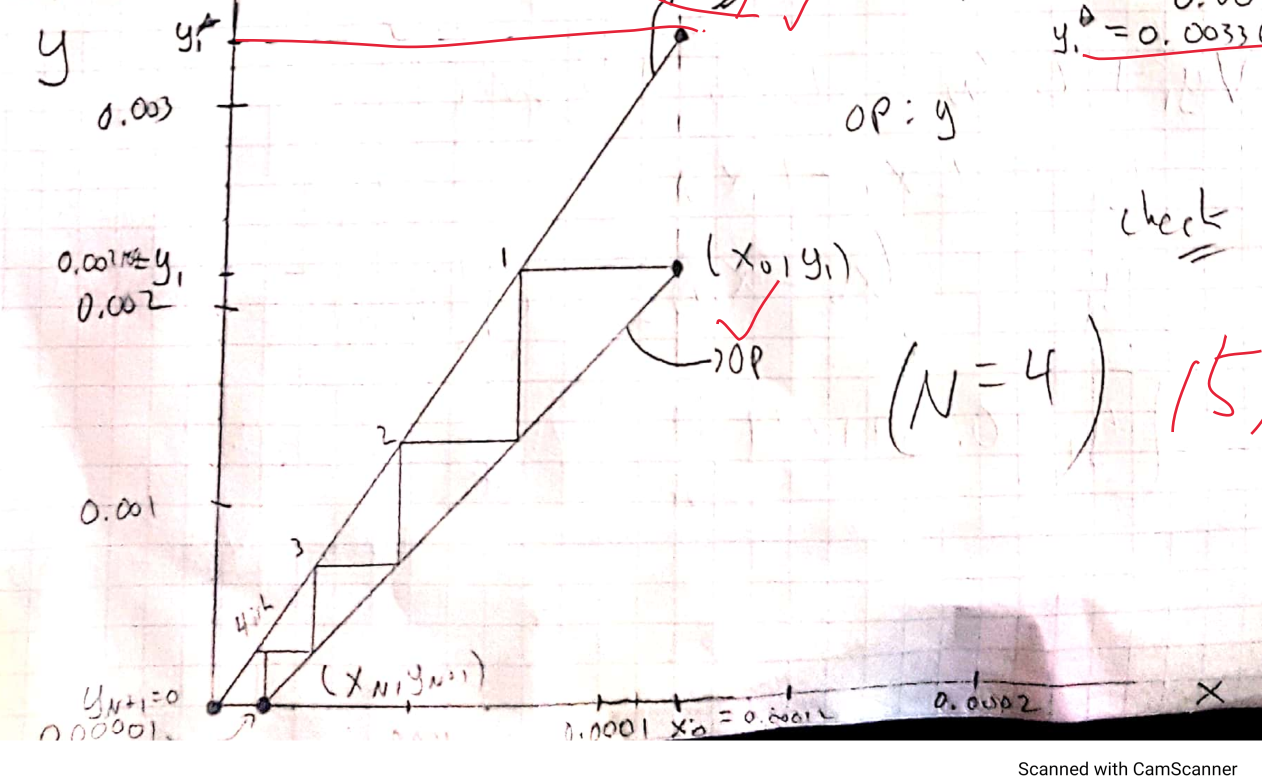
$= 28(0.0000108$

$(y^*_{N+1} = 3.024\times10^{-4})$

$(y_1^* = 0.00336)$

$L/V = 10$

C. McCabe-Thiele



Eq: $y = 28x$

$y_1^* = 28x_0 = 28(0.00012)$

$= 0.00336$

$y_1^* = 0.00336$

OP: y

check

$(N = 4)$

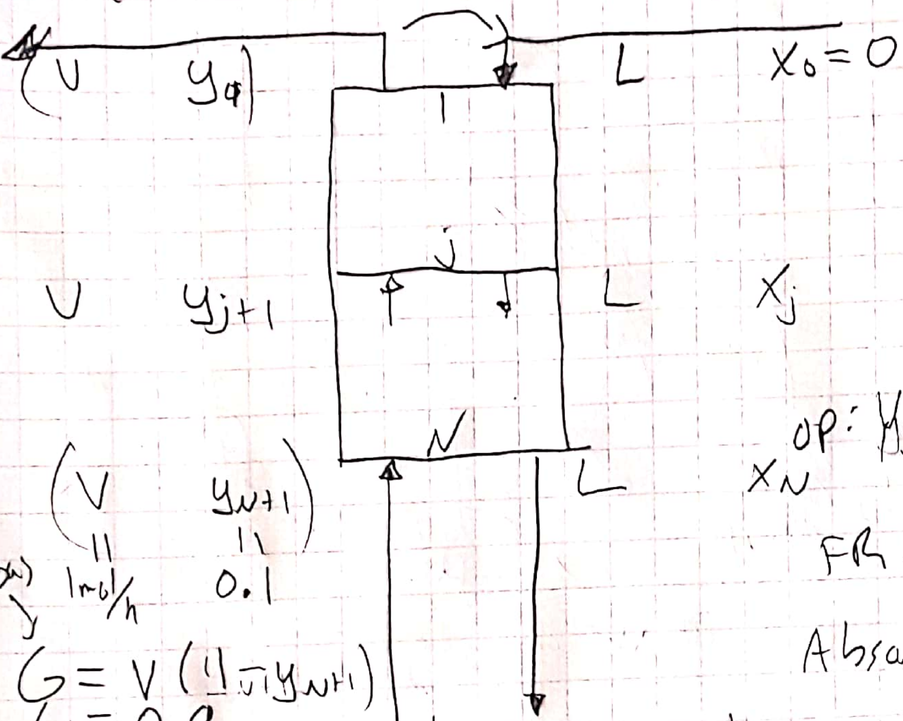
15/15

b) Draw ...

②

2) $\underline{\text{Absorber}}$ mole $\underline{\underline{\text{Concentrated}}}$ X Y

$(1-FR_{NB})$ Absorber pure water



$S/G = 45$

$y = 2.8x$

op: $Y_{j+1} = \frac{S}{G} X_j + (Y_+ - \frac{S}{6} X_0)$

(w) $V = 1\,mol/h$, $y_{N+1} = 0.1$

$G = V(1-y_{N+1})$
$G = 0.9$

$FR = 0.95$

Absorb 95% NO

a) Find (X_0, Y_1) (X_N, Y_{N+1}) $X = \frac{x}{1-x}$

$Y = \frac{y}{1-y}$

(1) $G = V(1-y_{N+1})$ (insoluble carrier)
$= (1)(1-0.1)$
$(G = 0.9\,kmol/h)$

(2) $S/G = 45 \Rightarrow S = 45G = 45(0.9) = 40.5\,mol/h$
$(S = 40.5\,mol/h) = (L)?$ (L Not constant Concentrated)

$[S = (1-x_N) L] \Rightarrow L = \frac{S}{1-x_N}$

(3) $\underline{FR}$ $(1-FR_{NB}) = \frac{\cancel{V} y_1}{\cancel{V} y_{N+1}}$

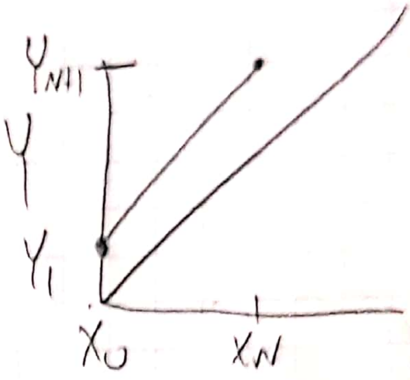
$y_1 = (1-FR_{NB}) y_{N+1} = (1-0.95)(0.1)$
$(\underline{y_1} = 0.005)$

MB $\quad Vy_1 + LX_N = Vy_{N+1} + \cancel{LX_0}$

$$LX_N = Vy_{N+1} - Vy_1$$

$$X_N = \frac{Vy_{N+1} - Vy_1}{L} = \frac{V(y_{N+1} - y_1)}{S/(1-X_N)}$$

$$\frac{S}{1-X_N}(X_N) = V(y_{N+1} - y_1)$$



$$\left(X_0 = \frac{x_0}{1-x_0} = \frac{0}{1} = 0\right)$$

$$\left(Y_{N+1} = \frac{y_{N+1}}{1-y_{N+1}} = \frac{0.1}{1-0.1} = 0.11111\right)$$

$$\left(Y_1 = \frac{y_1}{1-y_1} = \frac{0.005}{1-0.005} = 0.005025\right)$$

Point to put

OP

$$Y_{j+1} = (5/6)X_j + [Y_1 - (5/6)X_0]$$

X_N ↳ $Y_{N+1} = (5/6)X_N + [Y_1 - (5/6)\cancel{X_0}]$

$$Y_{N+1} - Y_1 = \frac{5}{6}X_N$$

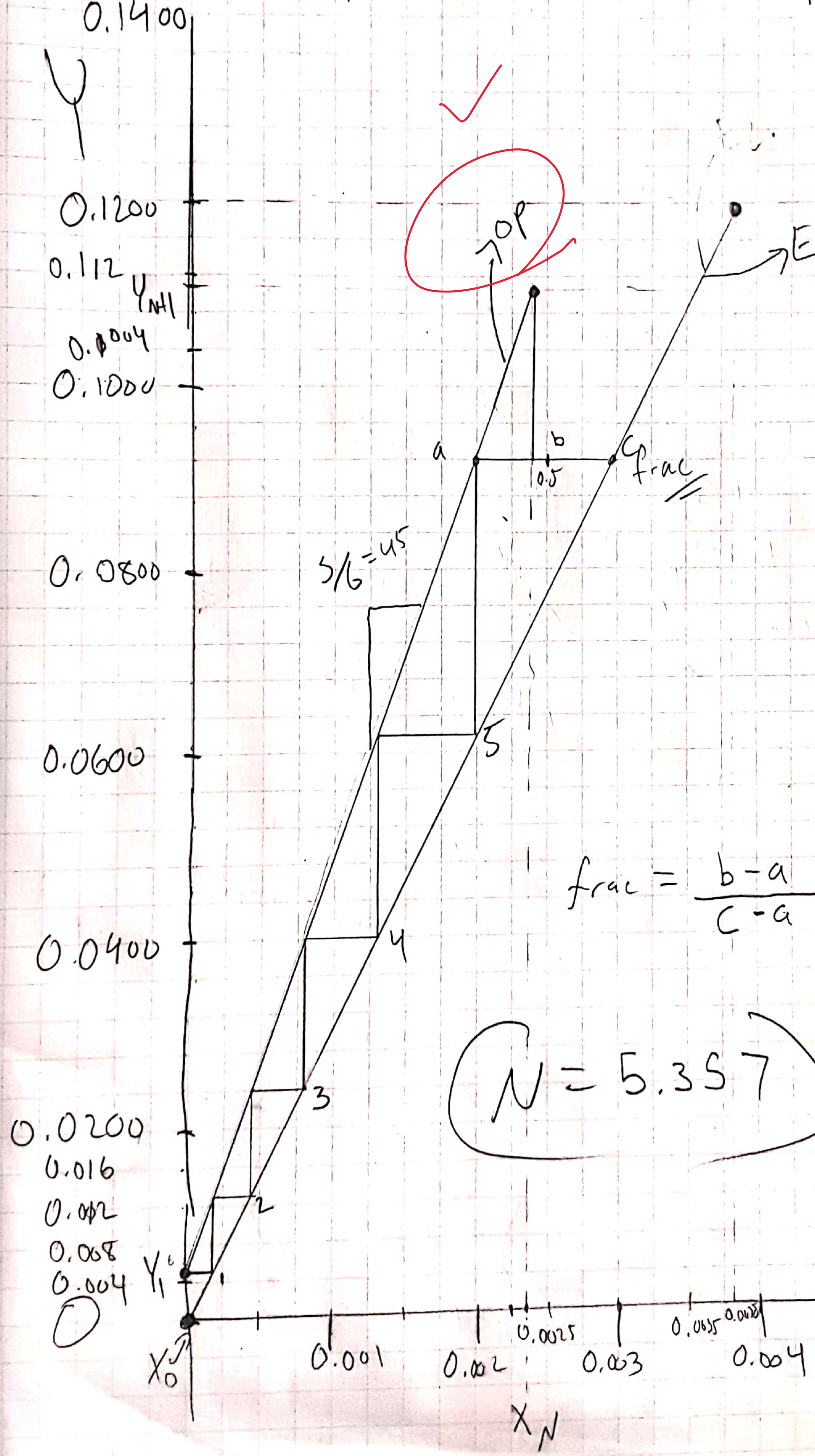
$$X_N = \frac{Y_{N+1} - Y_1}{(5/6)} = \frac{0.1111 - 0.005025}{45}$$

20/20

$$X_N = 0.002357$$

$$\Rightarrow \boxed{\begin{array}{l}(X_0, Y_1) = (0, 0.005025) \\ (X_N, Y_{N+1}) = (0.002357, 0.111111)\end{array}}$$

b) Draw OP line + Find N Absorber



check

$$5/6 = \frac{y_{N+1} - y_1}{X_N - X_0}$$

$$45 = \frac{0.1111 - 0.0\ldots}{0.002357\ldots}$$

$$\boxed{45 \approx 45}$$

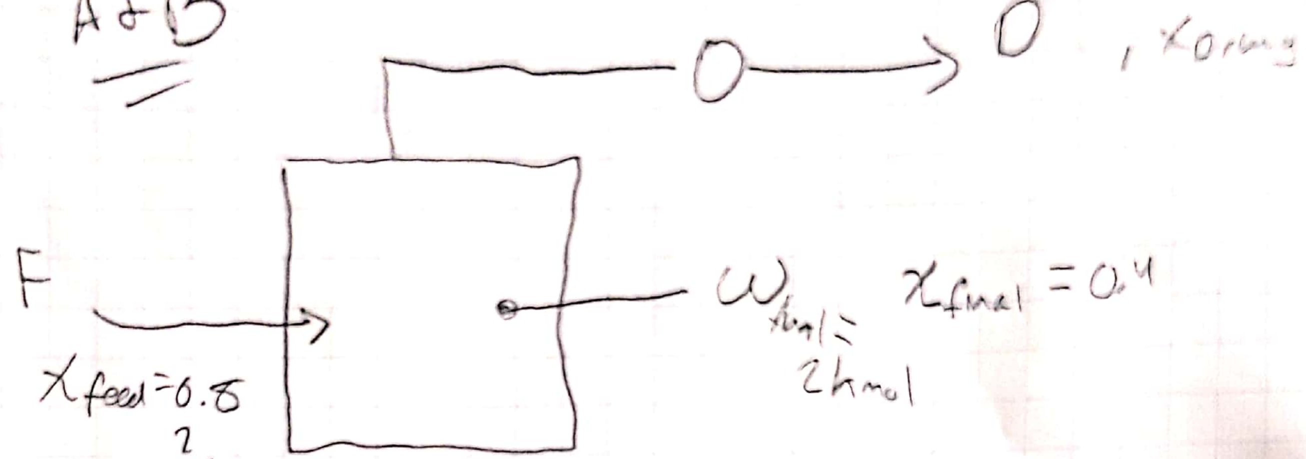
$$frac = \frac{b-a}{c-a} = \frac{0.002357 - 0.002}{0.003 - 0.002} = 0.357$$

$N = 5.357$

20/20

3) Simple batch

A + B



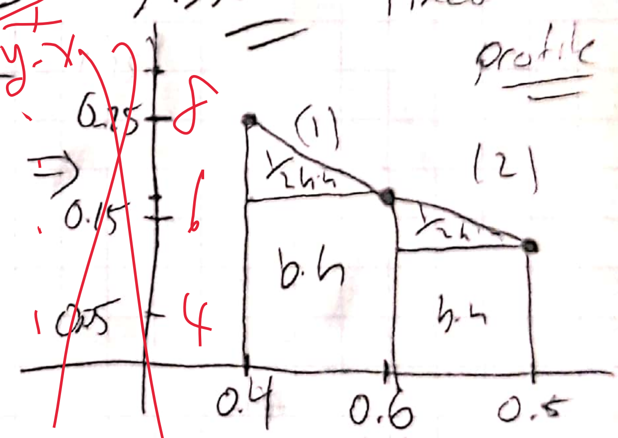
MB: $F = W_{final} + D_{tot}$

(1) Raleigh: $W_{final} = F\left(-\underbrace{\int_{x_{fin}}^{x_{feed}} \frac{dx}{y-x}}_{Area}\right)$

$$\left(F = -\frac{W_{final}}{Area}\right)$$

Break: $Area = \int_{x_{final}}^{x_{feed}} \frac{dx}{y-x}$ — Assume linear profile

x	y	y − x	$\frac{1}{y-x}$
0.2	0.7	0.5	
$\boxed{0.4}$	0.65	$\boxed{0.25}$	4
0.6	0.767	$\boxed{0.167}$	6
$\boxed{0.8}$	0.925	$\boxed{0.125}$	8
1.0	1	0	



$$Area = \left[(0.2)(0.167) + \left(\tfrac{1}{2}\right)(0.2)(0.25-0.167)\right]^{(1)} + \left[(0.2)(0.125) + \left(\tfrac{1}{2}\right)(0.2)(0.167-0.125)\right]^{(2)}$$

$(Area = 0.0417 + 0.0292 = \cancel{0.0709})$ 2.4

$$\left(F = -\frac{W_{final}}{Area} = \frac{2\,kmol}{\cancel{0.0709}} = 28.208\,kmol\right)$$

exp(−Area)

$D_{tot} = F - W_{formal}$

$= 28.208 - 2$

$D_{tot} = \cancel{26.208}$ kmol

20.04 kmol

10/20

Scanned with CamScanner